

1 Introduction and Theory

In many real-world environments, absolute reward signals are unavailable, highly noisy, or uninformative. Instead, relative preference feedback—knowing only that option A is better than option B—is far more reliable. **Dueling Bandits** formalize this problem by querying pairs of arms and receiving binary preference feedback.

This project proposes applying a Contextual Dueling Bandit framework to historical ATP match data from 2008 to 2023 [1]. The objective is to define, then identify the “best” tennis player under varying environmental contexts (court surfaces) and evaluate how 2 fundamental bandit algorithms handle the sudden introduction of new arms (a new generation of players).

1.1 Dueling Bandits

In a conference in 2009, Yue et al. proposed a new online optimization problem called the *K-armed Dueling Bandits Problem* where they studied a ‘partial-information online-learning game between pairs of strategies’ [4]. Eye doctors use pairwise duels (‘Option 1 or Option 2’) to converge on your optimal prescription; by determining the winner of each duel, the doctor is able to ascertain the optimal candidate (lens).

1.2 Measures of Regret and Optimality

Because the complete historical preference matrix P can be calculated offline directly from the dataset, we have access to the ground-truth probabilities for evaluation. We will measure:

- **Cumulative Strong Regret (R_T):**

$$R_T = \sum_{t=1}^T (\Delta_{i^*,i} + \Delta_{i^*,j}) \quad (1)$$

where $\Delta_{i^*,i} = P(i^* \succ i) - 0.5$. This measures how much “loss” the algorithm incurs by failing to select the true best player i^* for its duels.

- **Condorcet Winner** The **Condorcet winner** is the logical best result from a head to head style tournament. Unfortunately, while this allows for very nice regret analysis, it isn’t guaranteed to occur and in fact, commonly fails to exist in the real world, as shown in Section 6.1 of [5].
- **von Neumann Winner** As shown in [2], the **von Neumann winner** is the minmax or maxmin winner and is guaranteed for every game by von Neumann’s minmax theorem. The von Neumann winner also has the excellent property of being equal to the Condorcet winner in games where such an action exists.
- **Borda Winner** The Borda score s_i for player i is the sum of the probabilities of i winning against all other players in the set $\mathcal{P}$: $s_i = \sum_{j \in \mathcal{P}, j \neq i} P(i \succ j)$. Then the **Borda winner** is then defined as $i^* = \arg \max_{i \in \mathcal{P}} s_i$. This criterion is often preferable for the following reasons. First, a Borda winner is guaranteed to exist in any finite set of players, even when Condorcet cycles exist. Furthermore, while a Condorcet winner must defeat *every* individual opponent, the Borda winner represents the player with the highest average performance across the entire field, making it representative of overall dominance and still interpretable. Finally, in the dueling bandit setting, algorithms targeting the Borda winner, similar to Borda-UCB, exhibit lower regret and sample complexity compared to those seeking a Condorcet winner [3].

2 Methodology

Our problem requires the underlying preference graph of arms to be highly connected (i.e., the players must have played against each other historically). To satisfy this constraint, the arm pool was restricted to two overlapping generations; **The Veterans** include established top players active since the late 2000s (e.g., Federer, Nadal, Djokovic, Murray). **The ‘Next Generation’** of players who debuted or broke into the top tier around 2016 and have a dense history of matches against the veterans (e.g., Zverev, Medvedev, Tsitsipas).

To check the viability of the project, we computed an offline preference matrix across all 23 players and their matches across all surfaces. When checking for a Condorcet winner on each surface, we encountered a Condorcet cycle on Grass and Hard. A Condorcet cycle can be thought of as an extension of the rock-paper-scissors game. Federer may beat Nadal, who beats Djokovic, who beats Federer. This forces us to pivot to a different target criteria, namely the Borda winner for its properties described in Section 1.2, but we may still compare our Borda winner to the Condorcet winner where it exists.

We evaluate two distinct approaches to the dueling bandit problem. The first, **Borda-UCB**, is a context-free algorithm that maintains empirical estimates of each player’s Borda score ($\hat{b}_i$), selecting players based on a standard Upper Confidence Bound: $U_i = \hat{b}_i + \alpha\sqrt{\ln t/N_i}$. This provides a baseline for exploration-exploitation without considering environmental factors. The second, **Pairwise LinUCB**, incorporates context by modeling the preference probability $P(i \succ j)$ as a linear function of anti-symmetric feature vectors $\phi_{i,j} \in \mathbb{R}^d$, which include surface-specific win rates, ATP rank differences, and recent form. By utilizing a ridge regression estimate $\hat{\theta} = A^{-1}b$, the algorithm selects the arm that maximizes the aggregated Borda-UCB score across the active pool $\mathcal{A}_t$, allowing for faster adaptation to surface-specific mastery and the “moving target” dynamics of the professional tour.

3 Environmental Design

In the original implementation of our algorithm, we did not force the algorithms to pick sequential games. So if they wanted to see Federer play Nadal on Grass, we gave them a random sample from all their Grass matches. This was problematic for a few reasons, mainly that we were breaking assumptions of the Dueling Bandits game, but it also prevented the algorithms from learning about players as their career matured. This led to a classic ML issue (leakage) since we were using the player’s average rank across their career. Once the mistake was recognized, we updated the code to create a sequential queue for each pair, then forced the algorithms to pop the top match from the queue, picking their first match and getting more recent matches as time went on. Under our first implementation, we had 2 contextual datapoints; the difference between player’s average ranks and whether it was a 3 or 5 game match. After the updates, we increased our context to 4 datapoints; the difference in rank between players, the best-of-three or best-of-five indicator, the comparative surface mastery, and the form difference. These were carefully calculated using data only from matches before the match in question.

The core experiment will test the algorithms’ ability to generalize and adapt to non-stationary arm pools (the ‘rookie’ introduction at τ) based on sequential matches.

4 Results

The Rookie Injection Regret Analysis

In both the cases shown in Figure 1, Borda-UCB needs a period of acclimatization to the new arms while LinUCB simply treats them as new feature vectors, which it already has opinions about. This is seen best in the Clay example. LinUCB’s brief bump in regret for the Hard court relates to the quality of players who arrive in the Next Gen and their early and major upsets of established players. Meanwhile, both algorithms accurately identified the Condorcet Winner on Clay, Rafael Nadal, the ‘King of Clay’. They performed well against the true Borda results and our predictions, as can be seen in Table 1.

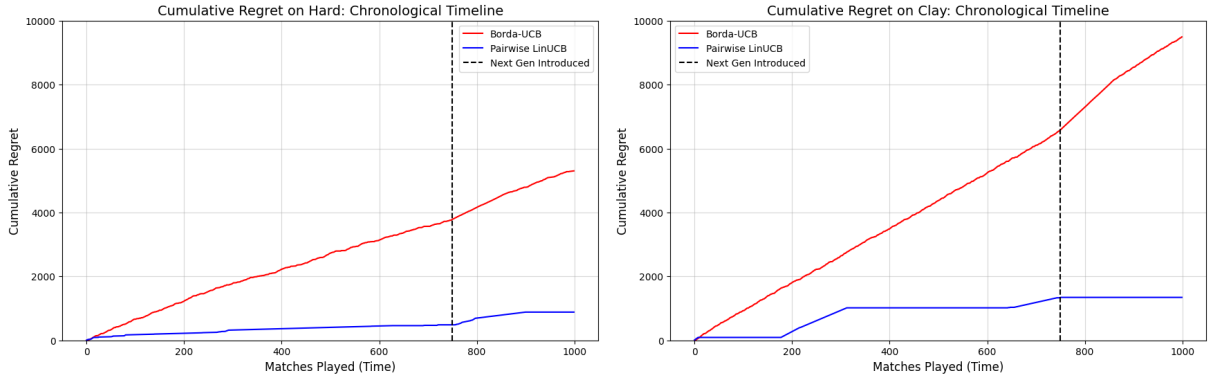


Figure 1: Regret for Hard and Clay court matches when introducing the ‘next gen’ at $T = 750$

Table 1: Comparison of Predicted and Actual Borda Win Probabilities

Model	Rank	Grass Court		Clay Court		Hard Court	
		Player	Prob.	Player	Prob.	Player	Prob.
Borda-UCB	1	Federer R.	0.605	Nadal R.	0.763	Djokovic N.	0.813
	2	Lopez F.	0.558	Djokovic N.	0.647	Federer R.	0.753
	3	Djokovic N.	0.549	Wawrinka S.	0.591	Nadal R.	0.729
LinUCB	1	Federer R.	0.513	Nadal R.	0.585	Djokovic N.	0.642
	2	Nadal R.	0.510	Djokovic N.	0.550	Kyrgios N.	0.585
	3	Murray A.	0.509	Federer R.	0.543	Berdych T.	0.560
Truth	1	Federer R.	0.602	Nadal R.	0.868	Djokovic N.	0.787
	2	Murray A.	0.568	Djokovic N.	0.714	Federer R.	0.769
	3	Djokovic N.	0.436	Federer R.	0.574	Nadal R.	0.701

5 Conclusion and Future Extensions

This study demonstrated that professional tennis matches can be effectively modeled as a dynamic dueling bandit problem using the Borda success criterion. While context-free Borda-UCB proved robust in stable environments, the Pairwise LinUCB algorithm, when properly augmented with useful context, successfully addressed the ‘moving target’ problem inherent in non-stationary sports data. Our results confirm that incorporating contextual features allows for faster adaptation to surface specialists and performance streaks, reducing cumulative regret during the injection of new players.

Interesting Extensions

1. Deeper regret bound analysis for the above problem
2. Test on other algorithms (Beat-the-Mean, Relative Confidence Sampling, SAVAGE, etc)
3. Different Paradigms, for example, algorithm sparring from [2].

References

- [1] Dissfya. Atp tennis 2000-2023 (daily pull). <https://www.kaggle.com/datasets/dissfya/atp-tennis-2000-2023daily-pull>, 2023. Accessed: 2026-03-01.
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- [5] M. Zoghi, S. A. Whiteson, M. de Rijke, and R. Munos. Relative confidence sampling for efficient on-line ranker evaluation. In *Proceedings of the 7th ACM International Conference on Web Search and Data Mining*, page 73–82. Association for Computing Machinery, 2014.

A Code, LLM Chats, and Algorithm Descriptions

Github code

Gemini used for project refinement and brainstorming

Gemini used for debugging LaTeX and code cleanup

Algorithm 1 Dynamic Borda-UCB

Require: Players $\mathcal{P}$, Injection time T_1 , Initial size K , Exploration α

- 1: **for** $t = 1, \dots, T$ **do**
 - 2: Define active pool $\mathcal{A}_t \leftarrow \{1 \dots K\}$ if $t < T_1$ else $\mathcal{P}$
 - 3: Select player $i_t = \arg \max_{i \in \mathcal{A}_t} \left(\hat{b}_i + \alpha \sqrt{\frac{\ln t}{N_i}} \right)$ *(Play $N_i = 0$ first)*
 - 4: Sample random opponent $j_t \sim U(\mathcal{A}_t \setminus \{i_t\})$
 - 5: Observe duel outcome $r_t \in \{0, 1\}$ for player i_t
 - 6: Update empirical Borda score $\hat{b}_{i_t}$ and pull count N_{i_t}
 - 7: **end for**
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Algorithm 2 Dynamic Pairwise LinUCB

Require: Players $\mathcal{P}$, Injection time T_1 , Exploration α

- 1: Initialize $A = I$ and $b = \mathbf{0}$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: Define active pool $\mathcal{A}_t \leftarrow \{1 \dots K\}$ if $t < T_1$ else $\mathcal{P}$
 - 4: Observe context, form anti-symmetric features $\phi_{i,j}$ for all $i, j \in \mathcal{A}_t$
 - 5: Estimate weights $\hat{\theta} = A^{-1}b$
 - 6: Compute pairwise UCB: $u_{i,j} = \phi_{i,j}^\top \hat{\theta} + \alpha \sqrt{\phi_{i,j}^\top A^{-1} \phi_{i,j}}$
 - 7: Select main player $i_t = \arg \max_{i \in \mathcal{A}_t} \frac{1}{|\mathcal{A}_t| - 1} \sum_{j \neq i} u_{i,j}$
 - 8: Sample random opponent $j_t \sim U(\mathcal{A}_t \setminus \{i_t\})$
 - 9: Observe duel outcome $r_t \in \{0, 1\}$ and played feature $x_t = \phi_{i_t, j_t}$
 - 10: Update statistics: $A \leftarrow A + x_t x_t^\top$ and $b \leftarrow b + (r_t - 0.5)x_t$
 - 11: **end for**
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